AES differential discovery laboratory

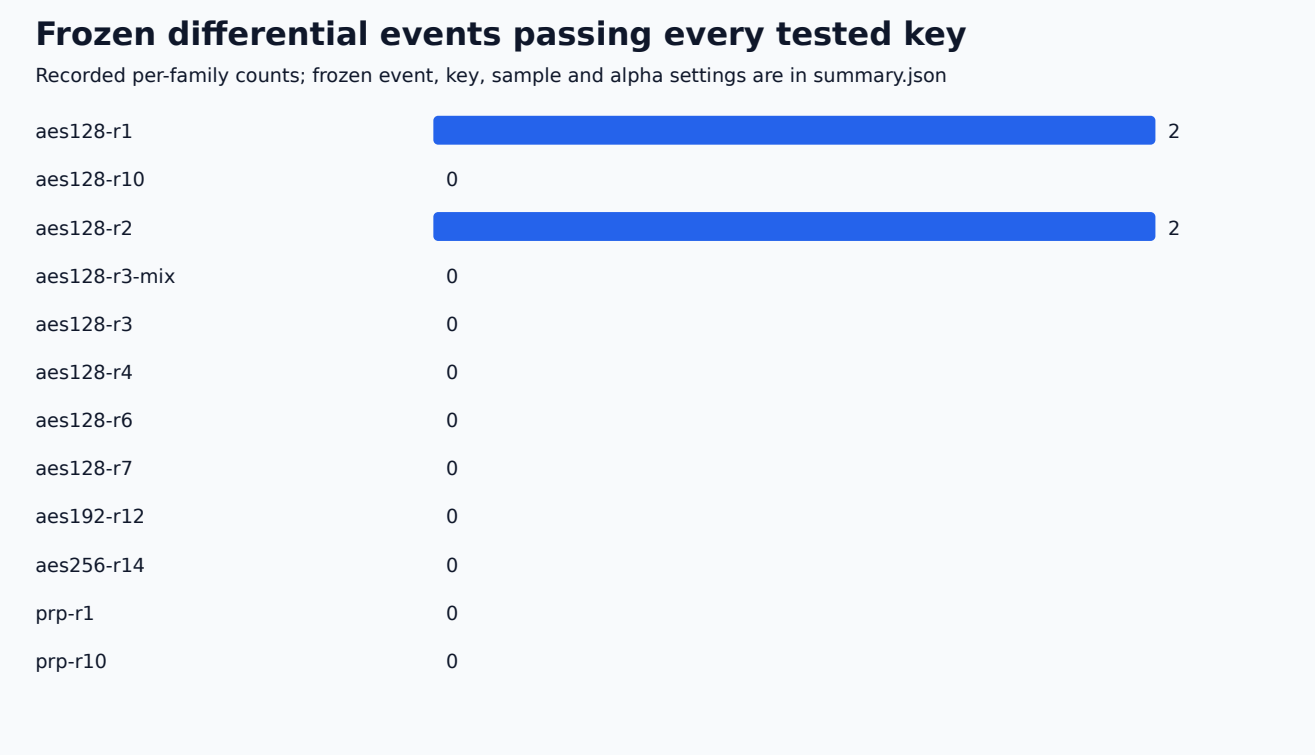
Descriptive rendering of recorded synthetic fixtures

41/41 recorded commands exited successfully. Recorded source revision: 8a3ee1d85cbb32b0fb78aa8b92cc363e3e0caffd. The raw directory retains exact commands, seeds, executable/source/protocol hashes, logs and statuses.

Differential endpoint observations

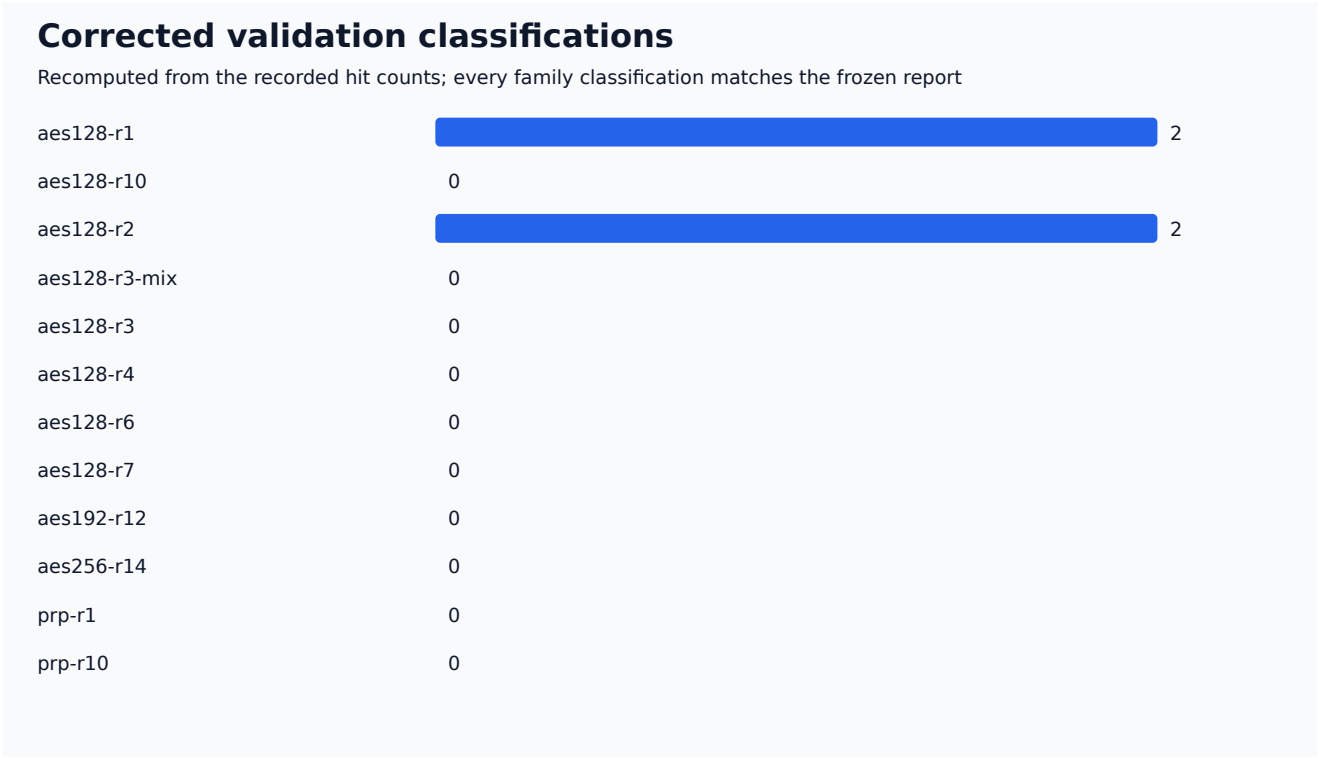
12 recorded differential families; 2 families contain an event passing every tested key. Families with validated events: aes128-r1, aes128-r2. All matched controls passed: true. The sum of recorded per-family alpha allocations is 0.006000. This allocation is meaningful only when frozen before execution; retrospective collection does not create a campaign error guarantee.

Per-family frozen event counts, keys, sample counts, input differences and alpha are in summary.json. Source JSON preserves the bounds calculated by its recorded revision; check that implementation before interpreting them as current bounds. A failed matched control withholds that family’s candidates.



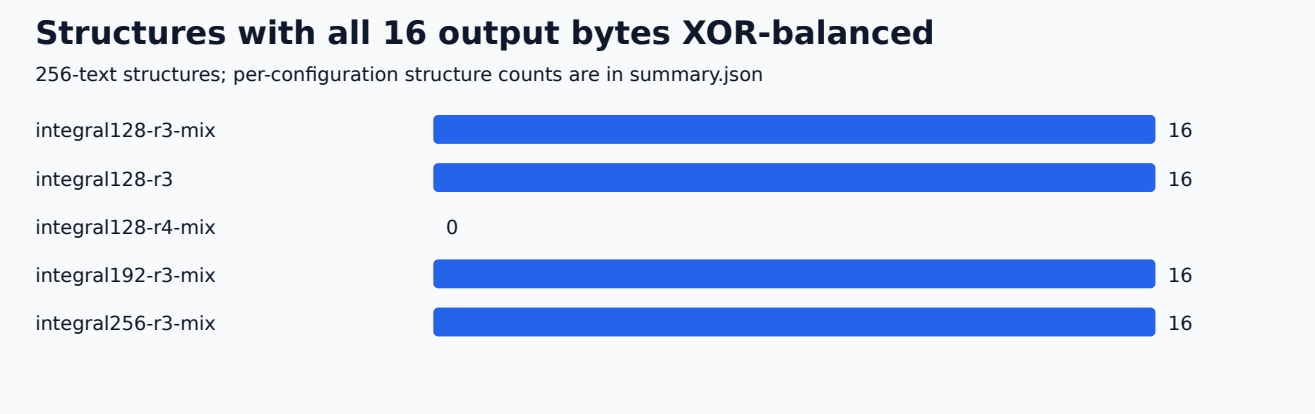
Corrected statistical interpretation

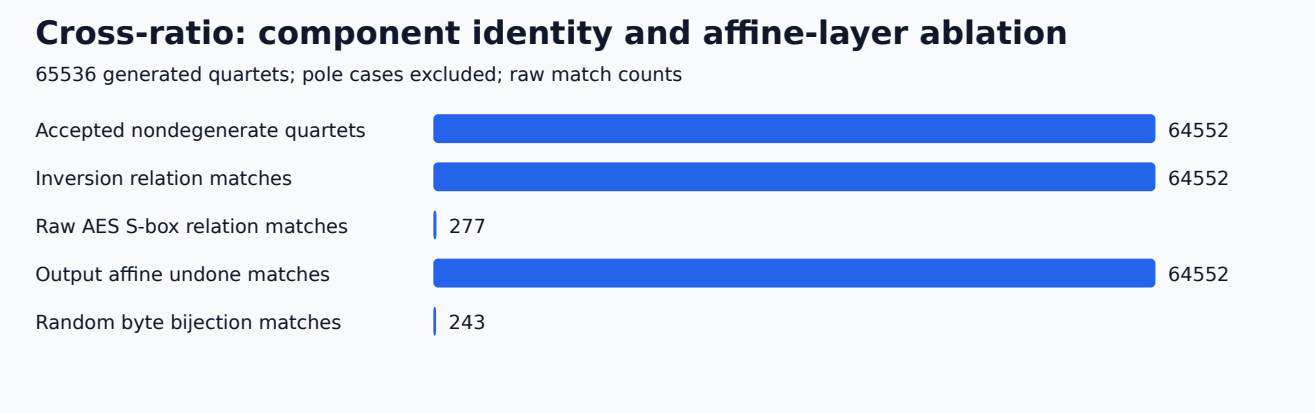
The archived primary/control bound fields retain the original two-sided shortcut and are not conservative bounds. The count-only reanalysis in ../../statistical-recheck.json is bound by SHA-256 in summary.json; it recomputes separate lower/upper Chernoff tails from archived hit counts without new cipher calls or samples. Every family classification is unchanged.



This does not establish security, absence of a different distinguisher, novelty, key recovery, or universal key independence. All per-key counts and the original recorded bounds remain in the source JSON.

Integral and component observations





Of 64552 accepted nondegenerate quartets, inversion matches: 64552; raw AES S-box matches: 277; affine-undone matches: 64552; random byte-bijection matches: 243. These are local component observations, not an invariant through an AES round or a reproduction of a complete published attack.

Source for the component experiment: https://eprint.iacr.org/2026/1549.

Validation and limits

The native summarizer derives only descriptive counts from recorded results. It does not independently verify the truth of every receipt. Summary JSON retains source-artifact hashes. Deterministic replay is not independent scientific review. No throughput, attack-cost gain or scientific campaign promotion follows.